

Element C: Presentation and Justification of Solution Design Requirements

Our school's students need more time, space, and equipment, because there isn't enough time to workout in the weight room with the limited amount of time given. The problem is that all of the solutions that we have come up with, either take a faculty member's time or the school's money to create. The people involved include Mr. Braxton, he is the athletic director. MSA students, these are the kids who use the gym. Lastly, our school as a whole, this is due to the money that would be needed and space.

Our goal is to give students more opportunity to exercise while attending strength and conditioning. The current situation is that two groups must have enough time to get their workout in from group 3:45 pm-5:15pm. The first group of students have from 3:45-4:30 (underclassmen). The second group is from 4:30-5:15 (upperclassmen). This limits the time that the students have. Although, if underclassmen ask to stay throughout that period they are usually allowed.

Below are our prioritized design requirements:

1. More active time exercising, more than 45 minutes of exercise (high)
2. Safe for students and coaches (high)
3. Doesn't require additional coaches or time (high)
4. Additional gym equipment in the weight room (medium)
5. More students can actively participate, currently 60 students participate (low)

Our idea is to survey students who have participated in strength and conditioning in the past. We hope to gain information from these interviews. This information will help lead us to understand the students perspective on the weight room. We showed the students our idea on separating the group into three sections, this will provide them with

the opportunity to exercise more. They will be given a schedule in which they would lift in the weight room for 30 minutes, do aerobic exercise for 30 minutes, and finally do 30 minutes of optional bone density exercises such as jump roping, lunges, pushup, etc.

Our three highest priority design requirements is to give students a more productive time in the gym. Students could be given a training plan, this would help with efficiency but this may need to be optional, some students may already have a plan for what they wish to do in the gym. There are some students who do not use their time productively in the gym. We think that it would also be beneficial for there to be a requirement of the amount of students who could participate in strength and conditioning.